Linux Handbook

1. INTRODUCTION TO LINUX

*Linux is a free, open-source Operating System kernel created by **Linus Torvalds** in 1991. It manages hardware and lets programs run on top of it.*

Linux Architecture

Hardware
↓
Kernel
↓
Shell (CLI)
↓
Applications / Utilities

Why Linux?

Key Reasons

- ★ Free & Open Source
- ★ Highly stable & secure
- ★ Preferred for servers, cloud & DevOps
- ★ Powerful command line control
- ★ Used in 90%+ of cloud servers

Kernel vs Operating System

Term	Meaning
Kernel	Core program that talks directly to hardware (CPU, memory, disk).
OS / Distro	Kernel + shell + utilities + package manager, packaged for users.

Popular Linux Distributions

Distro	Based On	Common Use
Ubuntu	Debian	Desktops, servers, beginners
CentOS / RHEL	Red Hat	Enterprise servers
Debian	-	Stability-focused servers
Fedora	Red Hat	Latest features, dev workstations
Kali Linux	Debian	Security & penetration testing
Amazon Linux	RHEL/Fedora	AWS EC2 instances

Good to Know

- ★ Linux is case-sensitive: **File.txt** ≠ **file.txt**
- ★ Everything in Linux is treated as a **file** (even devices!)
- ★ Shell is the interface between user and kernel (bash is most common)

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2. File System Hierarchy & Navigation

Linux organizes everything under a single root directory **/** — unlike Windows there are no C:, D: drives.

File System Hierarchy

Path	Purpose
/	Root of entire filesystem
/bin, /usr/bin	Essential user binaries/commands
/etc	System configuration files
/home	Personal user directories
/var	Logs, spool, variable data
/tmp	Temporary files (cleared on reboot)
/root	Home directory of root user
/dev	Device files

Path Types

Absolute vs Relative

Absolute: starts from root — **/home/aman/notes**

Relative: starts from current dir — **../notes** or **./file.txt**

Navigation Commands

Command	Description	Example
pwd	Print current working directory	pwd
ls	List files/folders	ls -la
cd	Change directory	cd /var/log
cd ..	Move one directory up	cd ..
cd ~	Go to home directory	cd ~
tree	Show directory structure as tree	tree -L 2

Useful ls flags

- **-l** → long listing (permissions, owner, size)
- **-a** → show hidden files (starting with .)
- **-h** → human-readable sizes (with -l)
- **-t** → sort by modified time

Tips

★ Use **Tab** key for auto-completion of paths
★ **cd -** takes you back to the previous directory
★ Hidden files/folders start with a dot, e.g. **.bashrc**

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3. FILE & DIRECTORY OPERATIONS

These commands let you create, copy, move, rename and delete files & directories.

Core Commands

Command	Description	Example
touch	Create empty file	touch app.log
mkdir	Create directory	mkdir project
mkdir -p	Create nested directories	mkdir -p a/b/c
cp	Copy file/folder	cp file.txt backup/
cp -r	Copy directory recursively	cp -r src/ dest/
mv	Move or rename	mv old.txt new.txt
rm	Delete file	rm file.txt
rm -r	Delete directory recursively	rm -r folder/
rm -rf	Force delete, no prompts	rm -rf temp/
rmdir	Delete empty directory	rmdir empty_folder

Finding Files

Command	Use
find . -name '*.log'	Find by filename pattern
find / -type d -name logs	Find directories named logs
locate filename	Fast search using index
which python3	Find path of a command

Caution

Be Careful With

- ★ **rm -rf /** can wipe the whole system!
- ★ There is **no Recycle Bin** in Linux CLI
- ★ Always double-check path before rm -rf

Example

```
$ mkdir -p project/src
$ cd project/src
$ touch index.js
$ cp index.js ../backup_index.js
# copies index.js one level up as backup_index.js
```

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4. FILE PERMISSIONS & OWNERSHIP

*Every file/folder in Linux has an **owner**, a **group**, and a set of read/write/execute permissions for each.*

Permission String Format

```
-  rwx  rwx  rwx
|  |    |    |
type owner group others
r=read(4)  w=write(2)  x=execute(1)
```

Numeric (chmod) Values

Value	Permission	Symbol
7	read+write+execute	rwx
6	read+write	rw-
5	read+execute	r-x
4	read only	r--
0	no permission	---

Common Commands

Command	Meaning
chmod 755 file	rwx for owner, r-x for group/others
chmod +x script.sh	add execute permission
chown user file	change file owner
chown user:group file	change owner & group
chgrp group file	change group only

Example

```
$ ls -l deploy.sh
-rwxr--r--  1 aman  devs  120  deploy.sh

$ chmod 750 deploy.sh    # owner: rwx, group: r-x, others: none
$ sudo chown aman:devs deploy.sh
```

Tips

★ First character in **ls -l** shows type: **d**=directory, **-**=file, **l**=symlink
★ Only owner or root can change file permissions
★ Use **chmod -R** to apply recursively to folders

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5. Viewing & Editing Files

Linux offers multiple ways to view file content directly in the terminal, plus terminal-based editors for making changes.

Viewing Commands

Command	Description
cat file.txt	Print entire file content
less file.txt	View file page-by-page (scrollable)
more file.txt	Similar to less, simpler
head -n 10 file	Show first 10 lines
tail -n 10 file	Show last 10 lines
tail -f file.log	Live-follow a growing log file

Nano (Beginner Friendly)

Nano Shortcuts

Ctrl+O	Save file
Ctrl+X	Exit editor
Ctrl+K	Cut line
Ctrl+U	Paste line

Vim (Powerful, Modal)

Vim Basics

i	Enter Insert mode
Esc	Back to Normal mode
:wq	Save & quit
:q!	Quit without saving
dd	Delete current line
yy / p	Copy / paste line

Example

```
$ tail -f /var/log/syslog
# keeps printing new lines as they get added — great for live logs
$ nano server.conf
# opens config file in nano to edit
```

Tips

- ★ Use **less** for large files — it doesn't load the whole file into memory
- ★ Press **q** to quit **less** / **more**
- ★ **vim** has a steep learning curve but is extremely fast once learned

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6. TEXT PROCESSING & FILTERS

Linux shines at chaining small tools together using pipes to search, filter and transform text.

Search & Filter

Command	Description	Example
grep 'text' file	Search lines containing text	grep 'error' app.log
grep -i	Case-insensitive search	grep -i 'ERROR' app.log
grep -r	Recursive search in folder	grep -r 'TODO' src/
grep -c	Count matching lines	grep -c 'fail' app.log
cut -d',' -f2	Extract a column (delimiter based)	cut -d',' -f2 data.csv
sort	Sort lines	sort names.txt
uniq	Remove duplicate adjacent lines	sort file | uniq
wc -l	Count lines in a file	wc -l file.txt

Redirection & Pipes

command > file (overwrite)

↓

command >> file (append)

↓

command < file (input from)

↓

cmd1 | cmd2 (pipe output)

sed / awk (quick use)

One-liners

sed 's/old/new/g'	Replace text
sed -n '5,10p' file	Print lines 5-10
awk '{print $1}'	Print 1st column
awk -F',' '{print $2}'	Custom delimiter

Example (chaining commands)

```
$ cat access.log | grep '500' | wc -l
# counts how many requests returned HTTP 500

$ ps aux | grep node | awk '{print $2}'
# prints PID of all running node processes
```

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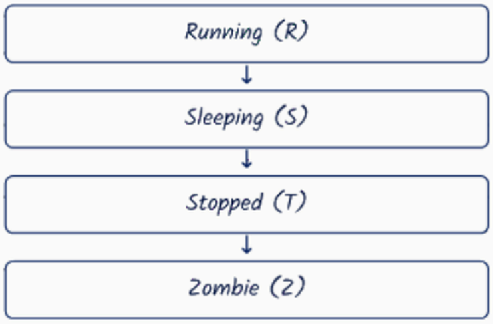
7. PROCESS MANAGEMENT

*Every running program is a **process** with a unique Process ID (PID). Linux gives fine control over starting, monitoring and stopping them.*

Viewing Processes

Command	Use
ps aux	List all running processes
top	Live process monitor (CPU/RAM)
htop	Improved, colorful version of top
pgrep name	Find PID by process name

Process States



Killing & Signals

Signal	Number	Meaning
SIGHUP	1	Reload/restart process
SIGINT	2	Interrupt (Ctrl+C)
SIGKILL	9	Force kill (cannot be ignored)
SIGTERM	15	Graceful terminate (default)

Job Control

- **command &** → run in background
- **jobs** → list background jobs
- **fg / bg** → bring job to foreground/background
- **nohup command &** → keep running after terminal closes
- **nice -n 10 command** → run with lower priority

Tips

★ **kill -9 PID** should be the last resort (no cleanup happens)
★ **kill PID** alone sends SIGTERM (graceful)
★ Use **top** then press **q** to exit

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8. USER & GROUP MANAGEMENT

Linux is a multi-user system. Each user belongs to one or more groups, and permissions/access are controlled at the user & group level.

User Commands

Command	Description
useradd -m username	Create new user with home directory
passwd username	Set/change user password
usermod -aG group user	Add user to a group
userdel -r username	Delete user & their home dir
whoami	Show current logged-in user
id username	Show UID, GID & groups

Group Commands

Command	Description
groupadd devs	Create new group
groups username	List groups a user belongs to
gpasswd -d user group	Remove user from group

Key Files

System Files

/etc/passwd	User account info
/etc/shadow	Encrypted passwords
/etc/group	Group definitions

sudo vs su

Command	Meaning
sudo command	Run a single command as root (with your password)
su -	Switch entirely to root user shell
sudo -i	Open a root login shell

Tips

- ★ Users with sudo access are usually in the **sudo** or **wheel** group
- ★ Never share the root password — prefer sudo for auditability
- ★ Use **id** to quickly check your own permissions

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9. PACKAGE MANAGEMENT

Package managers install, update and remove software along with their dependencies, so you rarely compile from source.

Debian/Ubuntu (APT) vs RedHat/CentOS (YUM/DNF)

Task	Debian/Ubuntu (apt)	RHEL/CentOS (dnf/yum)
Update package list	apt update	dnf check-update
Upgrade packages	apt upgrade	dnf upgrade
Install package	apt install nginx	dnf install nginx
Remove package	apt remove nginx	dnf remove nginx
Search package	apt search nginx	dnf search nginx
List installed	apt list --installed	dnf list installed

Low-level Tools

dpkg & rpm

dpkg -i pkg.deb	Install .deb file
dpkg -l	List installed packages
rpm -ivh pkg.rpm	Install .rpm file
rpm -qa	List installed packages

Snap / Flatpak

Universal Packages

- ★ **snap install code** — works across distros
- ★ Sandboxed, auto-updating apps
- ★ Slightly larger download size

Example

```
$ sudo apt update && sudo apt upgrade -y
$ sudo apt install docker.io -y
# updates package index, upgrades system, then installs docker
```

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10. Networking Commands

Networking commands help test connectivity, inspect interfaces, and transfer files/data between machines.

Connectivity & Interfaces

Command	Description
ping google.com	Test connectivity to a host
ip a / ifconfig	Show network interfaces & IPs
ss -tulpn / netstat -tulpn	Show open ports & listening services
curl -I url	Fetch HTTP headers of a URL
wget url	Download a file from a URL
traceroute host	Show route/hops to a host
nslookup / dig domain	DNS lookup for a domain

Remote Access

ssh user@host

↓

Enter password / use key

↓

Remote shell session

↓

scp file user@host:/path

Firewall (UFW - Ubuntu)

Common UFW Commands

ufw status	Check firewall status
ufw allow 22	Allow SSH port
ufw enable	Turn on firewall

Example

```
$ ssh aman@192.168.1.10
$ scp report.pdf aman@192.168.1.10:/home/aman/
# securely copies report.pdf to the remote machine
```

Tips

★ SSH default port is 22 — always change it on production servers
★ Use SSH keys instead of passwords for stronger security
★ **curl -O url** downloads & saves file with original name

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II. DISK, MONITORING & SERVICES

Keep track of disk usage, memory, and manage background services (daemons) using systemd.

Disk & Memory

Command	Description
df -h	Disk space usage (human-readable)
du -sh folder/	Size of a specific folder
free -h	RAM & swap usage
mount /dev/sdb1 /mnt	Mount a filesystem
umount /mnt	Unmount a filesystem
lsblk	List block devices/partitions

systemctl (Services)

Command	Use
systemctl status nginx	Check service status
systemctl start/stop nginx	Start/stop a service
systemctl restart nginx	Restart a service
systemctl enable nginx	Auto-start on boot

Logs

journalctl

journalctl -u nginx Logs for a service
journalctl -f Live-follow system logs
journalctl --since today Today's logs

Tips

★ **df -h** shows filesystem-level usage; **du -sh** shows folder-level usage
★ A service that's **enabled** starts at boot; **active** means running right now
★ Use **htop** alongside **free -h** for a full picture